

CCSU BCA Semester 2 – Digital Electronics

2026 Exam-Oriented Important Topics + Short Notes

1. Number Systems

Short Note: Digital electronics uses binary representation. Practice decimal, binary, octal and hexadecimal conversions and binary arithmetic.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. Boolean Algebra

Short Note: Learn Boolean laws, De Morgan's theorems, SOP/POS and simplification.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Logic Gates

Short Note: Know AND, OR, NOT, NAND, NOR, XOR and XNOR truth tables. NAND and NOR are universal gates.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Combinational Circuits

Short Note: Prepare half/full adder, subtractor, encoder, decoder, multiplexer and demultiplexer with truth tables and basic applications.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Sequential Circuits

Short Note: Learn latches and flip-flops: SR, D, JK and T. Understand clocking and state changes.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. Counters & Registers

Short Note: Study asynchronous/synchronous counters and shift registers. Learn basic applications in counting and data movement.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. K-Map

Short Note: Use Karnaugh maps to simplify Boolean expressions. Practice grouping adjacent cells in powers of two and deriving simplified SOP/POS expressions.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **Number Systems** with suitable example/diagram/formula/program.
- Explain **Boolean Algebra** with suitable example/diagram/formula/program.
- Explain **Logic Gates** with suitable example/diagram/formula/program.
- Explain **Combinational Circuits** with suitable example/diagram/formula/program.
- Explain **Sequential Circuits** with suitable example/diagram/formula/program.
- Explain **Counters & Registers** with suitable example/diagram/formula/program.
- Explain **K-Map** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.